

Project focus: introduce the problem of segmenting brain ventricles over multiple MRI contrasts (multimodal approach) and implement a specific workflow aimed at evaluating how the resulting volumes are affected by noise characteristics.

Details:

1. Briefly review the topic of brain ventricles segmentation over MRI images.
2. Implement a workflow, motivating its main sub-steps, to pre-process an MRI image and segment brain ventricles based on methods we have seen during lesson.
3. Segment ventricles and calculate the respective cross-sectional area over axial slice number 16 in T1 volume.
4. Identify axial slices that contain the ventricles and extend the quantification of their cross-sectional area to the whole T1 volume. What are the main challenges of segmenting this lesion with respect to other cerebral tissues?
5. Repeat step 3 and 4 over the T2 volume. Are there main differences over the quantification of brain ventricles in T1 vs. T2 volumes?
6. Add noise to the original datasets and check the performances of your implemented workflow to different levels of noise.
7. [Optional] manually segment the ventricles starting from axial slice number 16, hence quantify segmentation performances in terms of sensitivity, specificity and Dice coefficient.

Note:

The implemented method does not need to be fully automated.

Dataset:

- `sub2_t1.nii` contains the T1 volume.
- `sub2_t2.nii` contains the T2 volume.
- use `niftiread` and `niftiinfo` functions to import the volumes and metadata.

Tips:

- Consider implementing your workflow over a ROI of the image.
- Useful tool `VolumeViewer` app (built-in in MATLAB).
- Useful functions `getpts`, `bwselect`, `imfill`.